

Generation Interconnection System Impact Study Report

For Queue Project Q17-0850

Washington County Cogeneration Facility 230 kV
December 03, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0850. The Interconnection Customer proposes a 388.3 MW solar photovoltaic facility interconnecting at the Washington County Cogeneration Facility 230 kV substation in the Ohio area. The request was received on August 22, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 350 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$46,365,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Ltg Pleasant Hill - Fedael Hts 100kV line: 145.3% loading (201 MVA) following the outage of the Barrett - Norwich 500kV Line; DFAX = 5.9%.

Ltg Pleasant Hill - Craigmont 100kV: 140.6% loading (217 MVA) following the outage of the Norwich - Barrett 500kV; DFAX = 5.9%.

The Washington County Cogeneration Facility 100/1 kV XFMR is overloaded to 124.4% of its 171 MVA emergency rating for the loss of the Washington County Cogeneration Facility - Mossyrock 230kV. The project contributes 19.1% DFAX (74.3 MW) to this overload.

Washington County Cogeneration Facility - Mossyrock 230kV L: 119.2% loading (928 MVA) following the outage of the Barrett - Wolfe 500kV line; DFAX = 53.0%.

Flatlick Substation - Combee Road 230kV: 102.6% loading (767 MVA) following the outage of the Omni-Hackberry - Norwich 500kV L; DFAX = 5.9%.

For the loss of Mossyrock - Washington County Cogeneration Facility 230kV Line, the Washington County Cogeneration Facility - Agua Mansa 100kV L loads to 101.8% of its 155 MVA rating. Project DFAX: 8.2%.

For the loss of Gavin Power Llc 500 kV/345 kV Auto, the Gavin Power Lc - Olympus 230kV loads to 101.7% of its 1,053 MVA rating. Project DFAX: 7.7%.

The Norwich - Larson Street 230kV is overloaded to 101.5% of its 1,113 MVA emergency rating for the loss of the Norwich - Barrett 500kV. The project contributes 14.0% DFAX (54.5 MW) to this overload.

Washington County Cogeneration Facility 230/1kV Transformer: 100.2% loading (213 MVA) following the outage of the Washington County Cogeneration Facility - Mossyrck 230kV; DFAX = 19.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ltg Pleasant Hill - Federal Hts 100kV L	Rebuild line with higher-capacity conductor	\$1,128,000	64% (\$719,000)
Ltg Pleasant Hill - Craigmont 100kV	Rebuild line with higher-capacity conductor	\$3,168,000	64% (\$2,019,000)
Washington County Cogeneration Facility 100/1 kV XFMR	Replace transformer with higher-rated unit	\$2,250,000	100% (\$2,250,000)
Washington County Cogeneration Facility - Mossyrock 230kV	Rebuild line with higher-capacity conductor	\$54,352,000	46% (\$24,718,000)
Flatlick Substation - Combee Road 230kV	Reconductor line	\$10,795,000	5% (\$521,000)
Agua Mansa - Washington County Cogeneration Facility 100kV L	Reconductor line	\$4,016,000	100% (\$4,016,000)
Gavin Power Lc - Olympus 230kV	Reconductor line	\$3,378,000	6% (\$210,000)
Larson Street Substation - Norwich 230kV	Reconductor line	\$4,865,000	8% (\$375,000)
Washington County Cogeneration Facility 230-1 kV Transformer	Replace transformer with higher-rated unit	\$11,537,000	100% (\$11,537,000)
	Total allocated to project		\$46,365,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.